MANACE INJECTION

DESCRIPTION: Clear, colourless liquid.

COMPOSITION: Each 1 ml ampoule contains Haloperidol 5 mg.

PHARMACODYNAMICS: It has been suggested that Haloperidol, like the phenothiazines, depresses the central nervous system at the subcortical level of the brain, the mid brain, and the brain-stem reticular formation. Haloperidol seems to inhibit the ascending reticular system of the brain-stem, possibly through the caudate nucleus, thereby interrupting the impulses between the diencephalon and the cortex. It may also inhibit uptake of neurohumoral amines in the mid-brain. Recently, it has been postulated that haloperidol acts by blocking dopaminergic neuro-transmission in the central nervous system.

PHARMACOKINETICS: Absorption, Metabolism and Excretion: Peak plasma levels of haloperidol occur within two to six hours of oral dosing and about twenty minutes after intramuscular administration. The mean plasma half-life (terminal elimination) has been determined as 20.7 ± 4.6 (SD) hours, and although excretion begins rapidly, only 24 to 60% of ingested radioactive drug is excreted (mainly as metabolites in urine; some in faeces) by the end of the first week, and very small but detectable levels of radioactivity persist in the blood and are excreted for several weeks after dosing. About 1% of the ingested dose is recovered unchanged in the urine. The slow excretion may be related to a high degree of plasma protein binding that in one study was observed to be 92%. Haloperidol is highly lipid-soluble and may remain in fatty tissue for some weeks.

INDICATIONS: Chronic Therapy: The management of manifestations of psychotic disorders such as schizophrenia, psychosis due to organic brain damage or mental deficiency, senile psychosis, the manic phase of manic depressive illness, Gilles de la Tourette syndrome.
Short Term Therapy: The treatment of acute alcoholism for the relief of delusions, hallucinations and confused states, and for the control of accompanying tremulousness and aggresiveness behaviour. In the treatment of intractable nausea and vomiting associated with radiation or malignancy and not responding to other therapy. Neuroleptanalgesia.

RECOMMENDED DOSAGE: Warning: Where rapid control of an acutely disturbed patient is required, or where heavier than usual dosages are envisaged, or when parenteral administration, particularly repeated parenteral administration is required, then the patient should be transferred as soon as possible to a situation where resuscitative measures and parenteral antiparkinson medication are available.
There is considerable variation from patient to patient in the amount of medication required for therapy. As with all antipsychotic drugs, it is important to titrate the dose of Haloperidol in accordance with the clinical effect and the severity of the disease. Monitoring of blood levels is not a routine procedure. Children and debilitated or geriatric patients may be more sensitive to Haloperidol, and the starting dose, maximum dose and maintenance doses are therefore generally lower for these patients. In all age groups, titration of dosage should be as rapid as possible. If therapy is commenced with parenteral Haloperidol, a change to oral medication should be made as soon as possible.
No pharmacokinetic data are available to enable special recommendations to be given for patients with renal or hepatic impairment. Also, no information on the effects of meals on drug absorption is available.
Parenteral Administration: Agitation and aggressiveness associated with acute psychosis: (eg. mania, hypomania, acute schizophrenia, toxic confusional states including delirium tremens), 2-10 mg IM or IV initially. The amount will depend on the patient's age, physical status, and severity of symptoms. The initial dose may be given as a slow IV injection or as a bolus. Depending on the response of the patient, subsequent doses may be given as often as half hourly for IV injections or hourly for IM injections. The total daily dose administered should not exceed 100 mg.
This treatment approach is not without risk and high doses of antipsychotic drugs should only be administered to the physically healthy adult.
Appropriate precautions should be taken in patients with a history or evidence of such conditions as cardiovascular disorders or epilepsy.
Maintenance phase: Once continuous therapeutic control is achieved, there is a need to determine maintenance dosage. As pharmacokinetic studies to date have not addressed this problem, it is necessary to make an estimate, assuming that plasma levels are to be maintained, that the plasma half life of haloperidol is approximately 24 hours, and that the initial control is attained within 24 hours.
The parenteral total daily maintenance dose: May be calculated as half the total daily dose used to achieve control. This dose can then be given in divided doses morning and evening. To avoid excessive plasma levels, giving of the first maintenance dose should be delayed if possible until 4-8 hours after the last controlling dose.

ROUTE OF ADMINISTRATION : i.m & i.v injection

CONTRAINDICATIONS: Comatose states: In the presence of central nervous system depression due to alcohol and other depressant drugs; Parkinson's disease; known hypersensitivity to haloperidol; and in patients with manifest lesions of the basal ganglia.
Use in paediatrics: As the safety and effectiveness of Haloperidol has not been generally established in children, the drug is not recommended for use in the paediatric age group except in severely aggressive and hostile children or for the treatment of the rare Gilles de la Tourette syndrome. Haloperidol should not be used in children under 3 years.

WARNING & PRECAUTIONS: Epilepsy: Haloperidol may be given to epileptics but normal anticonvulsant therapy should be continued as Haloperidol has no anticonvulsant activity. It has been reported that seizures can be triggered by Haloperidol in known epileptics that were previously controlled.
Thyrotoxicosis: Extrapyramidal side effects may be encountered at an early stage in the form of dystonic reactions or motor restlessness (akathisia). Thyrotoxic patients may be more prone to develop extrapyramidal symptoms.
Haloperidol may impair the mental and/or physical abilities required for the performance of hazardous tasks such as operating machinery or driving a motor vehicle. The ambulatory patient should be warned accordingly.
Elderly or debilitated patients receiving haloperidol should be observed for evidence of over-sedation which, unless alleviated, could result in complications such as terminal stasis pneumonia.
Elderly patients are more prone to orthostatic hypotension and show an increased sensitivity to the anticholinergic and sedative effects of haloperidol, and are also more prone to develop tardive dyskinesia and parkinsonism. Long term treatment may masks the symptoms of tardive dyskinesia. There is no known effective treatment but careful observation for early signs of tardive dyskinesia and reduction or discontinuation of dosage of medication may prevent a more severe manifestation of the syndrome.
Haloperidol does not usually affect blood pressure significantly but care should be exercised in patients with severe cardiovascular disorders or being treated with antihypertensive drugs because of the possibility of unexpected hypotension. Severe or prolonged hypotension may require vasopressors. Adrenaline should not be used since Haloperidol may block its vasopressor activity and cause further decrease in blood pressure.
Caution should be observed in patients with arteriosclerosis who may have occult lesions of the basal ganglia.
As with all antipsychotic agents, Haloperidol should not be used alone where depression is predominant. It may be combined with antidepressants to treat those indicated conditions where depression and psychosis coexist. When Haloperidol is used to control mania in cyclic disorders, a rapid mood swing to depression may occur.
It should be borne in mind that the antiemetic action of Haloperidol may relieve nausea and vomiting and so obscure the diagnosis of an underlying organic disorder which was causing these symptoms. Ocular or cutaneous changes and decreases in serum cholesterol have occurred following administration of a butyrophenone structurally related to Haloperidol. In the same study which reported these changes, Haloperidol was shown to be free of these side effects - however, it is advisable to carefully observe patients who receive Haloperidol for a prolonged period in order to identify any changes in the skin or eyes.
As with other antipsychotic agents, Haloperidol has been associated with rare cases of a neuroleptic malignant syndrome in which the patient experiences fever, profound diaphoresis, autonomic dysfunction, muscular rigidity, altered consciousness, and various levels of coma. Recovery is usually complete within five to seven days of discontinuing Haloperidol.
Rare cases of sudden and unexpected death have been reported in association with the administration of Haloperidol, although the nature of the evidence makes it impossible to determine the contributory role, if any, of the drug.
Sudden death, QT prolongation and Torsades de Pointes has been reported in patients treated with Haloperidol, especially when given intravenously.

INTERACTION WITH OTHER MEDICATIONS: Although Haloperidol has minimal soporific action it will potentiate the action of central nervous system depressants such as narcotics or other analgesics, barbiturates, or other sedatives, anaesthetics, tranquillizers or alcohol. However, Haloperidol may reduce the efficacy of anticonvulsants. Haloperidol may interfere with the activity of phenindione and coumarin anticoagulants.
Caution should be observed in the use of lithium salts together with high doses of Haloperidol; patients receiving combined therapy should be monitored closely for early evidence of

neurological toxicity and treatment discontinued promptly if such signs appear. An encephalopathy-like syndrome (characterized by weakness, lethargy, fever, tremulousness and confusion, extrapyramidal symptoms, leucocytosis, elevated serum enzymes, blood urea and fasting blood sugar) followed by irreversible brain damage has occurred in a few patients treated with lithium plus haloperidol. High doses of haloperidol may potentiate the action of methyldopa.
Beta-blockers - Concurrent use of Haloperidol and Propranolol resulted in severe hypotension or cardiopulmonary arrest on 3 occasions in a schizophrenic patient. The patient suffered apparent ill effects when the drugs were taken singly and at different times.
Concurrent use of antihistamines may intensify side effects especially confusion, hallucinations, nightmares and increased intraocular pressure because of secondary anticholinergic effects of haloperidol.
Paralytic ileus may occur with concurrent therapy. Also reduction of the antipsychotic effectiveness may occur due to the reduced gastrointestinal absorption.
Antimalarials - Increased risk of ventricular arrhythmias with halofantrine.

USE IN PREGNANCY & LACTATION: **Use in pregnancy:** Although there have been isolated case reports of birth defects, following foetal exposure to Haloperidol in combination with other drugs, no definite cause and effect relationship can be confirmed for butyrophenones. When given in high doses during late pregnancy, butyrophenones may cause prolonged extrapyramidal disturbances in the newborn infant.
Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalization.
Manace Injection should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.
Use in lactation: Safe use of Haloperidol by nursing mothers has not been established; therefore its use is not recommended.

SIDE EFFECTS: In the low dosage range (1-2 mg daily), adverse effects from Haloperidol have been infrequent, mild and transitory. In patients receiving higher doses some adverse effects are seen more frequently. Neurological effects are the most common.

Neurological Side Effects. As with all antipsychotic agents, extrapyramidal reactions such as Parkinson-like symptoms, akathisia or dystonic reactions, have been reported frequently. They are more commonly observed during the first few days of treatment but can also be seen at later stages.
The Parkinson-like symptoms are most common and when first observed are usually mild to moderately severe. They sometimes remit spontaneously as treatment continues, or can be relieved by a reduction in dose or the temporary use of anti-parkinson medication.
Other extrapyramidal reactions such as akathisia (motor restlessness) or dystonic, reactions (eg. hyperreflexia, opisthotonus, oculogyric crisis, dyskinesias) can be more severe. Akathisia is best managed by a reduction in dosage in conjunction with the temporary use of an oral anti-parkinson drug. Dystonias, which can produce laryngeal spasm or bronchospasm may be controlled by intravenous or intramuscular administration of benztropine mesylate (1-2 mg IM or IV) or biperiden (2-10 mg IM or IV), or intravenous diazepam (10 mg IV). A pseudo-parkinson rigidity syndrome may occur later during the course of treatment and may also respond to antiparkinson agents. Although the occurrence and severity of these reactions appear to be dose related over the moderate dosage range (2-15 mg) higher doses do not cause a significantly higher incidence of extrapyramidal reactions.
It has been reported that some extrapyramidal reactions may persist even after a reduction in dosage and/or treatment with antiparkinson drugs. In such cases the drug should be discontinued.
Persistent tardive dyskinesia: Tardive dyskinesia may appear in some patients on long-term therapy or may appear after drug therapy has been discontinued. The risk appears to be greater in elderly patients on high-dose therapy, especially females. The symptoms are persistent and in some patients appear to be irreversible. The syndrome is characterized by rhythmical involuntary movement of the tongue, face, mouth or jaw (eg. protrusion of tongue, puffing of cheeks, puckering of mouth, chewing movements). Sometimes these may be accompanied by involuntary movements of extremities. There is no known effective treatment for tardive dyskinesia: antiparkinson agents usually do not alleviate the symptoms of this syndrome. If these symptoms appear, it is suggested that the dosage of all antipsychotic agents be progressively reduced with a view of discontinuation if possible. Should it be necessary to reinstitute treatment or increase the dosage of the agent, or switch to a different antipsychotic agent, the syndrome may be masked. It has been suggested that fine vermicular movements of the tongue may be an early sign of the syndrome, and, if the medication is stopped at that time, the syndrome may not develop.
Other central nervous system effects: These are occasionally reported and include:- depression, anxiety, euphoria, lethargy, agitation, drowsiness, insomnia, headache, confusion, vertigo, apprehension, anorexia, grand mal seizures in previously controlled epileptics, apparent exacerbation of psychotic symptoms including hallucinations. Toxic psychosis may occur with overdosage.
Cardiovascular Effects: Tachycardia and orthostatic hypotension are occasionally reported. Increased respiratory rate has also been observed.
Endocrine Effects: Hyperprolactinaemia, lactation, gynaecomastia, menstrual irregularities, impotence or increased libido. Hypoglycaemia or hyperglycaemia are rare.
Miscellaneous: Haematologic effects have not been significant in patients receiving Haloperidol. There have been occasional reports of mild and usually transient leukopenia and leukocytosis, minor decreases in blood cell counts, anaemia or a tendency toward lymphomonocytosis. Agranulocytosis has only rarely been reported, and then in association with other medication.
Liver toxicity has been reported but a causal relationship has not been established.
Dermatologic effects are known to have occurred in isolated idiosyncratic patients - eg. maculopapular and acneiform skin reactions, photosensitivity and hair loss.
Other occasionally reported side effects are:- constipation, nausea, vomiting, dyspepsia, blurred vision, dry mouth, urinary retention, peripheral oedema, excessive perspiration or salivation, heartburn.

SYMPTOMS AND TREATMENT OF OVERDOSE: **Symptoms and Findings:** The manifestations would be an exaggeration of the known pharmacologic effects and adverse reactions. The most prominent symptoms would be:
1. Severe extrapyramidal reactions
2. Hypotension
3. Sedation

The patient would appear comatose with respiratory depression and hypotension which could be severe enough to produce a shock-like state. An extrapyramidal reaction would be manifest by muscular weakness or rigidity and a generalised or localised tremor. Hypertension rather than hypotension is also possible.
Treatment: There is no specific antidote. Treatment is largely supportive but gastric lavage or induction or emesis should be carried out (unless the patient is obtunded, comatose or convulsing), followed by administration of activated charcoal. cathartic may be administered.
For comatose patients, establish a patent airway by use of an oropharyngeal airway or endotracheal tube. Respiratory depression may necessitate artificial respiration.
Hypotension and circulatory collapse may be counteracted by use of intravenous fluids, plasma or concentrated albumin and vasopressor agents such as dopamine or noradrenaline. Adrenaline should not be used, since Haloperidol may reverse its action and cause profound hypotension.
In cases of severe extrapyramidal reactions, antiparkinson medication (eg. benztropine mesylate 1-2 mg IM or IV) should be administered parenterally. ECG and vital signs should be monitored. Hypothermia should be managed with external warming.

INCOMPATIBILITIES: Haloperidol Injection should not be mixed with other products unless their compatibility is known.

STORAGE CONDITIONS: Store below 30ºC. Protect from light. KEEP OUT OF REACH OF CHILDREN. *JAUHKAN DARIPADA KANAK-KANAK.*

PACK SIZE: Pack in 1 ml ampoules in a box of 10's.

SHELF LIFE: Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:
DUOPHARMA (M) SDN BHD
Lot 2599 Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA